Chap. 9
Senosr Noise

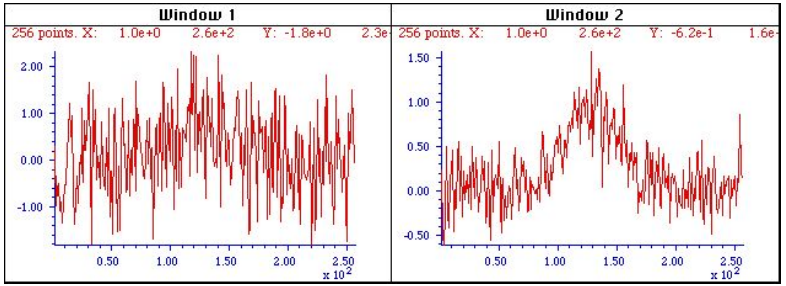
1. Noise and sensors
2. Extrinsic noise sources
3. Intrinsic noise sources
4. Guidelines to suppress noise

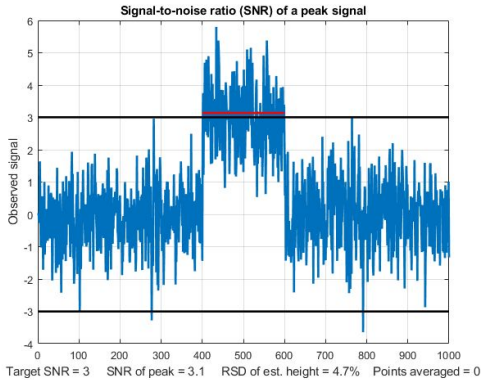
1. Noise and sensors – what is electrical noise?

- Noise is an unwanted additional signal which adds on the top of electrical signals from sensors or output devices.

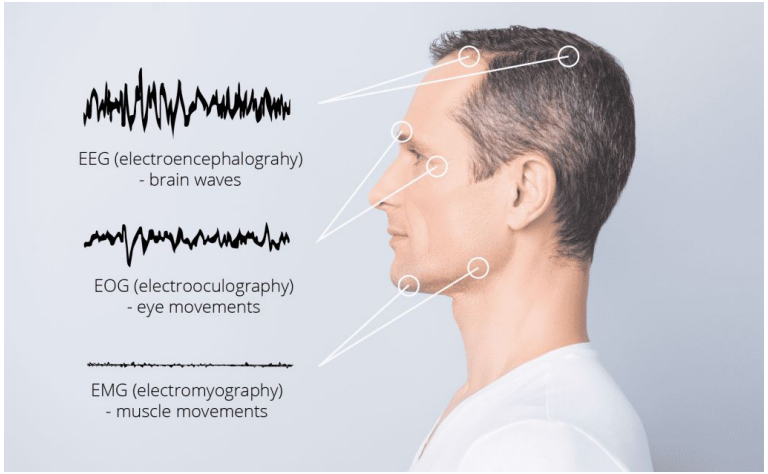
- Presence of noise could cause
  -- Digital signal: false “alarm” from LO to HI or vice versa
  -- analog signal: noise can “overwhelm” true small analog sensor signals.

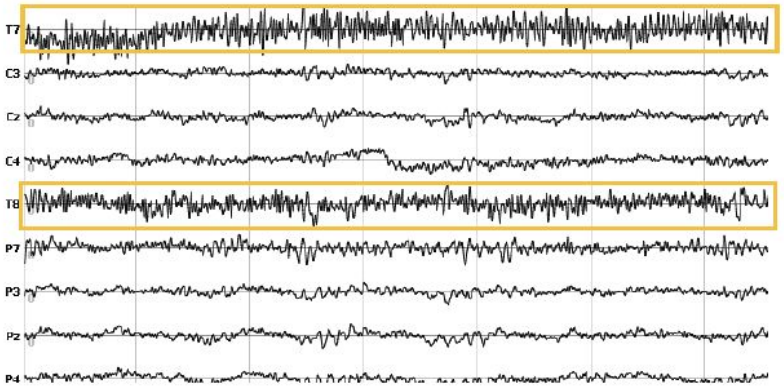
- In general, the amount of noise necessary to affect a digital signal is much higher because there is a large difference between HI (say, +5V) and LO (0 V) signals.

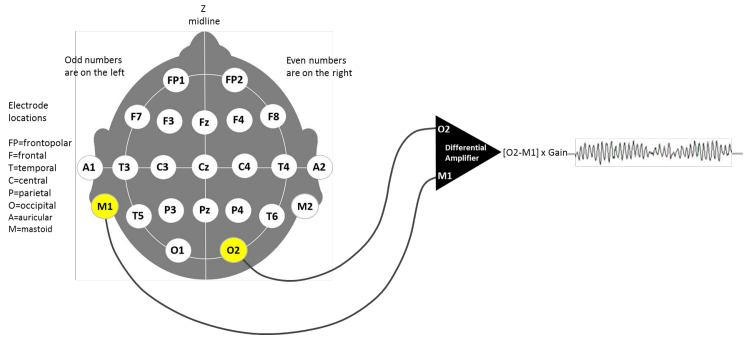




Example: noise in brain waves (EEG)







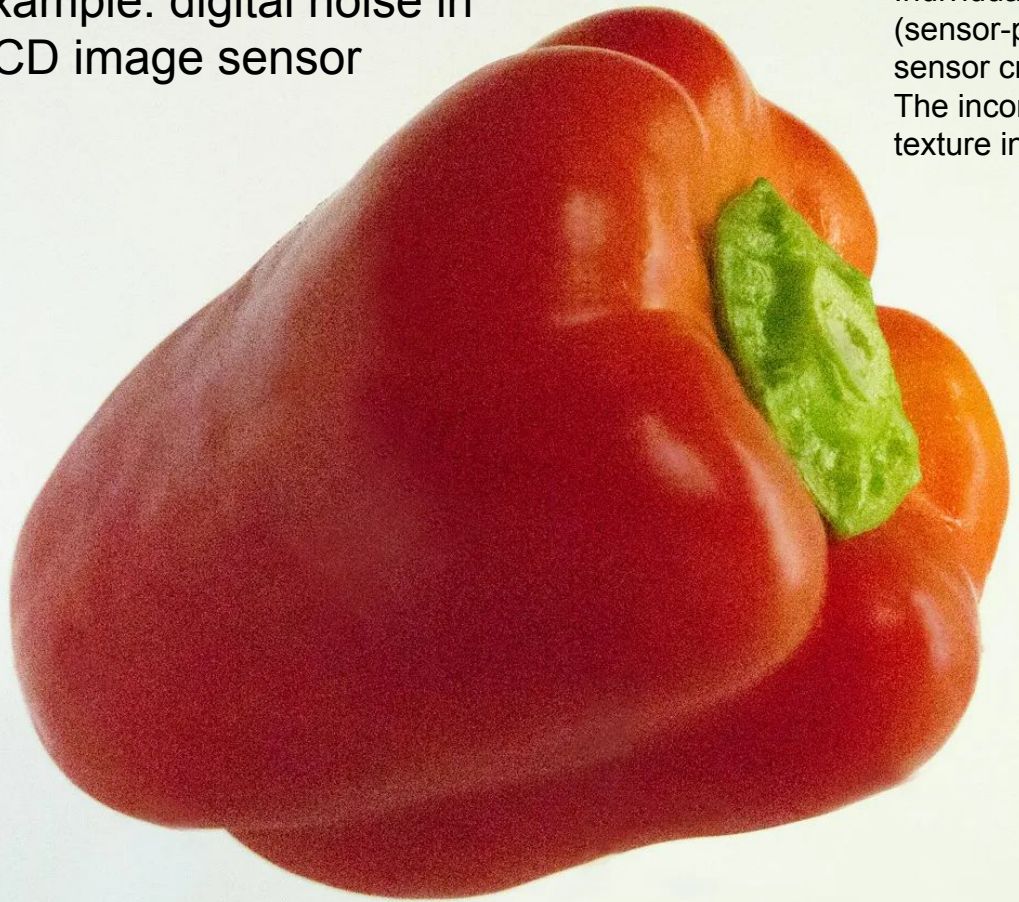
https://www.quora.com/What-is-the-meaning-of-O2-M1-in-EEG-data

Line noise from ac (50 Hz in the HK) can have strong artifacts on the electrode recording of raw EEG data. If the reference electrode is affected, the captured line noise appears in all other electrodes. The cognitive frequencies of the brain are often below the 50 or 60 Hz range. Line noise can be filtered out with electronic filters.

https://imotions.com/blog/eeg/

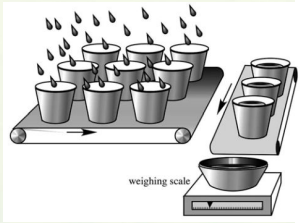
https://lsic.hkbu.edu.hk/en/facilities/mri.html
https://lsic.hkbu.edu.hk/

Example: digital noise in CCD image sensor



Individual sensor-points (sensor-pixels) on the digital image sensor create an incorrect data point . The incorrect point creates a grainy texture in the final picture.

CCD bucket analog



How to measure noise?

Signal-to-noise (S/N) ratio

Noise is measured by to signal-to-noise (S/N) ratio defined by:

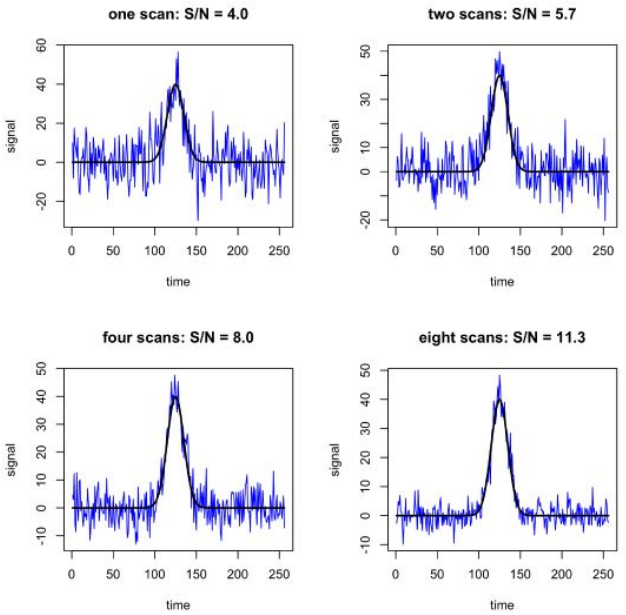
$$\text{SNR} = 10\log_{10}\frac{\text{signal power}}{\text{noise power}}$$

$$= 10\log_{10}\frac{v_{signal}^2}{v_{noise}^2} \quad \text{(unit: dB)}$$

v_{signal}^2: mean square voltage value of signal
v_{noise}^2: mean square voltage value of noise

e.g. v_{signal} = 1V (rms); v_{noise} = 0.1V (rms)

$$\text{SNR} = 10\log_{10}\frac{1^2}{0.1^2} \quad \text{(unit: dB)} = 20\text{dB}$$



The most important difference between the signal and the noise is:
signal is determinate (fixed in value);
noise is indeterminate (random in value).

By repeating measurements and doing averages (e.g. over time), noise will be reduced whereas signal will maintain its value – thus SNR will be enhanced.

Where does noise come from?

Extrinsic noise sources

Noise could develop in a sensor's output or its output circuitry– due to interactions with environment.

Intrinsic noise

Noise could originate from a sensor all by itself, even when it is completely isolated.

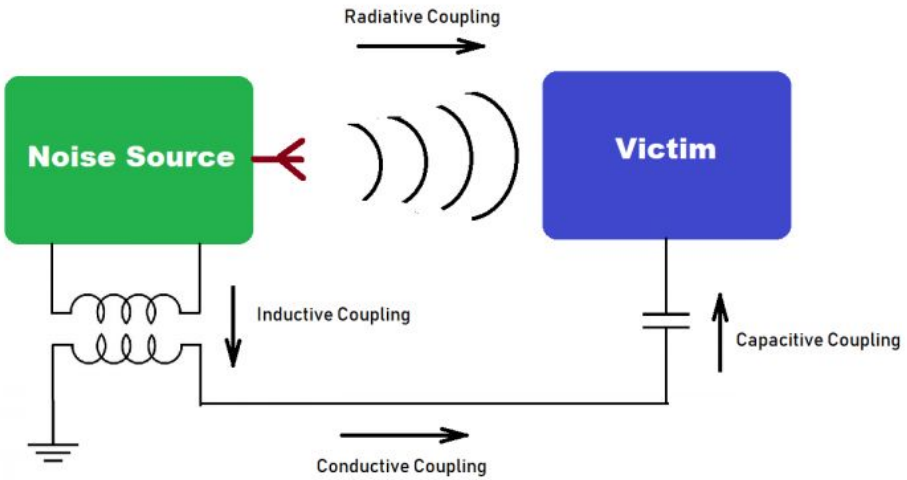
2. Extrinsic noise

Extrinsic noise

sources

- Sensor and its circuits cannot be totally isolated from environment.
- Noise comes from outside by:

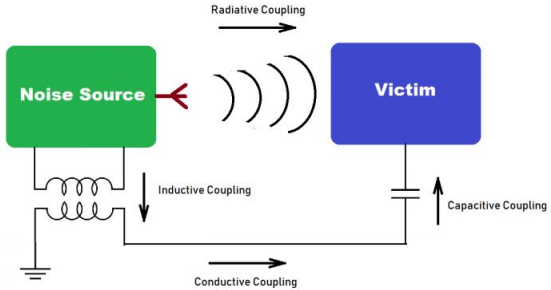
-- radiative coupling, e.g., spikes, atmospheric discharges
-- inductive coupling
-- capacitive coupling
-- conductive coupling (ground loops)

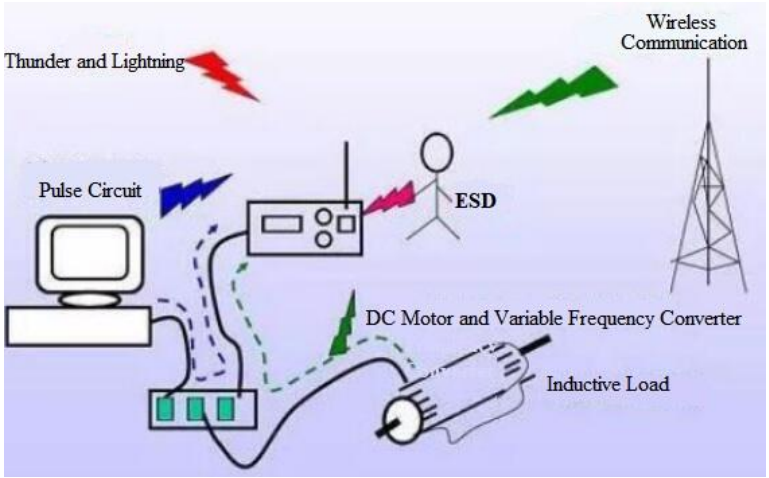


Source https://www.systoncable.com/how-to-reduce-electrical-noise/

Radiative coupling

- Radiative noise can come from high energy sources not in physical contact with the sensor circuits. The noise is just the result of propagation of electromagnetic (EM) waves disturbances through air or the ambient.
- Radiative noise is sometimes called EM interference or EMI.
- If radiative noise comes from ac supply, it is called “ac line noise” or 50 Hz noise (in HK)

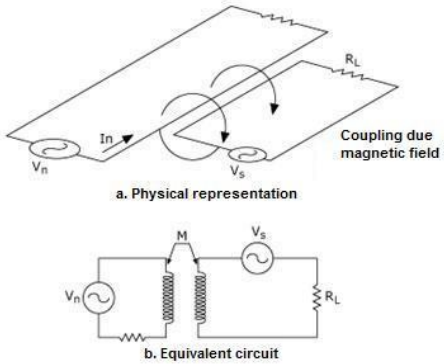


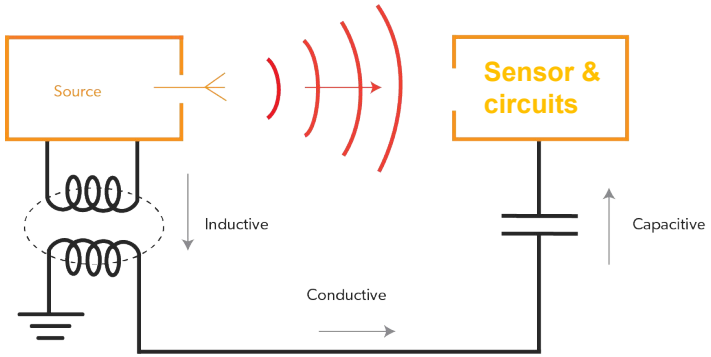


https://www.apogeeweb.net/article/137.html

Shielding:
make a metal case to enclose your sensor circuits

Inductive coupling (magnetic)





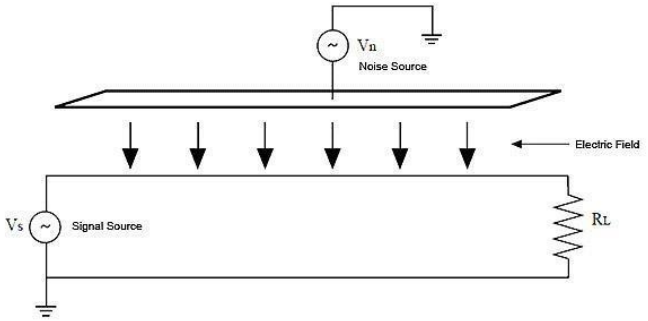
Noise could come from **magnetic coupling** from nearby current loops:

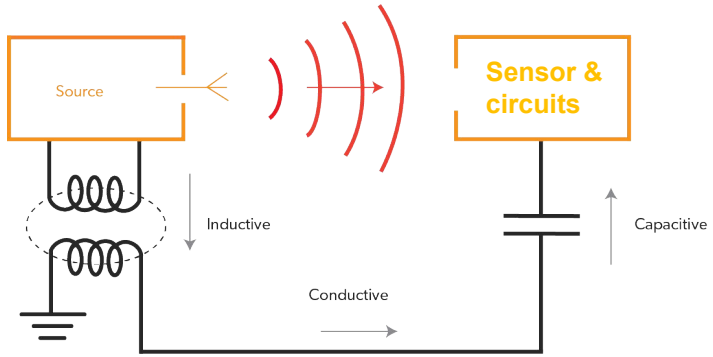
Noise will be reduced if the cable length in parallel is reduced

Short wires recommended

Twisted pair cables

Capacitive coupling (electric)





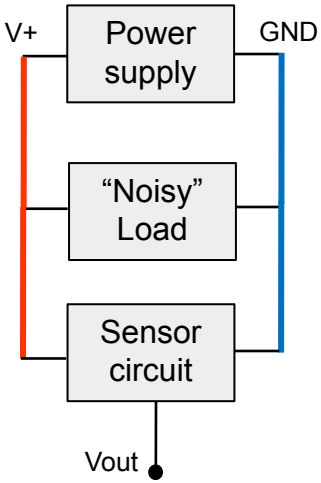
Noise could come from **electric coupling** from nearby charged objects:

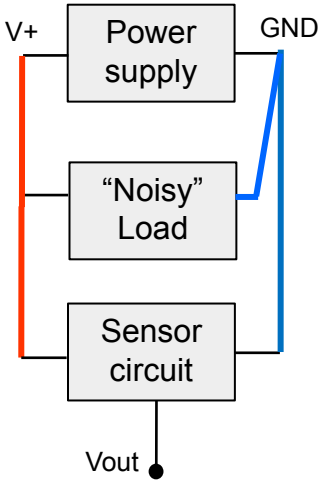
Capacitance will be reduced if the "plates" overlapping region is reduced.

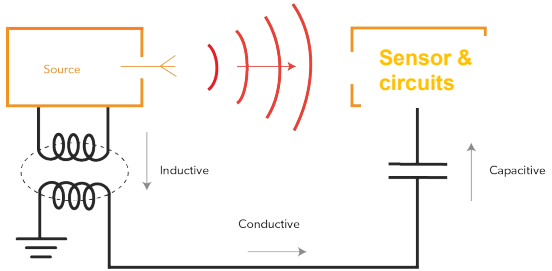
Short wires recommended

Shielded cables. Surround signal wires with grounded conductor.

Conductive coupling







<u>Conductive coupling</u>:

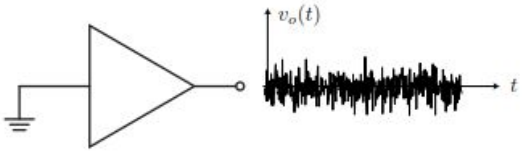
Conductive coupling can be reduced if short wires are used, and sensor makes direct connection to ground

Short wires recommended

Make direct connections to ground.

3. Intrinsic noise

- Even if all extrinsic noise sources are eliminated, circuits and electronic sensors still carry noise.
- This intrinsic noise originates from the particle nature of electrons. In a conductor, the conduction electrons behave as discrete particles, and they undergo random thermal motions at an absolute temperature T.
- Such random motion give rise to voltage or current fluctuation.
- For a resistor R, the thermal noise leads to a fluctuating noise voltage with a mean square value:



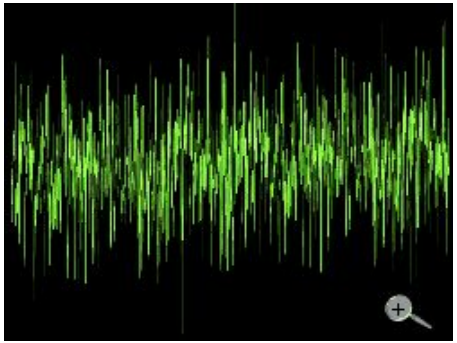
$$\overline{v_n^2} / \Delta f = 4kTR$$

k is the Boltzmann constant
$=1.38\times10^{-23}$ J/K
T = absolute temperature
Δf = freq. range in which noise is measured

Example:

R = 10kΩ
T = 20°C = 293K, 4kT = 1.62×10^{-20}J
Δf = 1 MHz
⇒

$$v_{n,rms} = \sqrt{\overline{v_n^2}} = 13\mu\text{V}$$



3. Intrinsic noise

THERMAL • thermal excitation of charge carriers



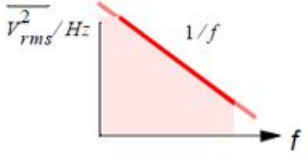
$$\frac{\overline{V_{rms}^2}}{\Delta f} = 4kTR \text{ , V}^2\text{/Hz}$$

SHOT • fluctuations in dc current flow through junctions



$$\frac{\overline{I_{rms}^2}}{\Delta f} = 2qI_{DC} \text{ , A}^2\text{/Hz}$$

FLICKER • traps in semiconductors



$$\frac{\overline{V_{rms}^2}}{\Delta f} = \frac{K_f \cdot V_{DC}^2}{f} \text{ , V}^2\text{/Hz}$$

T = Temperature, K
k = 1.38 x 10^{-23} J/K
q = 1.6 x 10^{-19} C
K_f = Flicker coefficient

https://tackbro.tistory.com/entry/Thermal-noise-Shot-noise-Flicker-noise

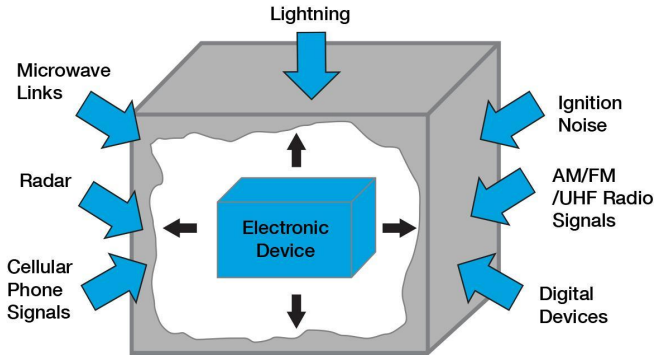
4. Guidelines to suppress electrical noise

- Identify the origin(s) of the noise sources.
- If you can, turn off these sources.

In reality, noise sources cannot be completely eliminated.
Measures that can be taken to suppress noise:

A. Shielding box

Enclose your sensor and sensor measuring system with a metal box. The box should be connected to a good conductor (usually grounded).

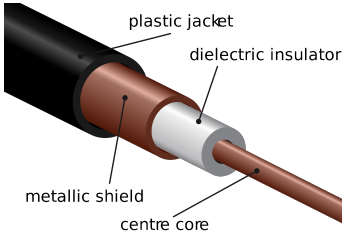


https://www.zatkoff.com/news/emi-shielding-materials-capabilities

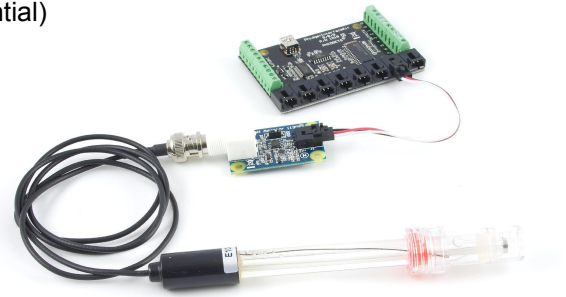
Measures that can be taken to suppress noise:

B. Use shielded cables to make connections

Similar in concept to a shielding box. A metallic shield (usu. at ground) encloses a signal wire in the core. The signal wire is then protected from noise interferences.



ORP sensor (oxidation/reduction potential)



https://www.phidgets.com/?tier=3&catid=11&pcid=9&prodid=414



Measures that can be taken to suppress noise:

C. Always use the shortest possible wires

With short wires, inductive, capacitive, and conducting coupling will be reduced.

Measures that can be taken to suppress noise:

D. Avoid ground loops

Sensors should make direct connections to the ground point. Do not “ride on” ground connections on other devices to reach ground.

